

Soo Min You

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Research Interests

Interested in advancing statistical methodology and machine learning for analyzing complex, high dimensional biomedical data, including time series, electronic health records, clinical studies, and other large scale health datasets.

Education

- 09.2024 – 05.2026 **Columbia University, New York, NY**
Master of Science in Biostatistics (expected)
- 09.2013 – 05.2017 **Cornell University, Ithaca, NY**
Bachelor of Science in Chemical Engineering

Research & Project Experience

- 02.2025 – Present **Research Assistant**, Department of Biostatistics, Columbia University, *New York, NY*
Advisor: Dr. Tian Gu
- Project 1:** Evaluating Representation Embeddings from LLMs and Time-Series Foundation Models for Wearable Accelerometer-Based Health Prediction.
Developed risk prediction models using embeddings derived from wearable accelerometer data and evaluated model performance using NHANES datasets.
Manuscript accepted to *AMIA 2026 Informatics Summit* (Co-first author)
- Project 2:** Spatial Transcriptomics in Cancer Biology
Joint project with Dr. Lulu Shang at MD Anderson Cancer Center.
Conducting a literature review and investigating multimodal integration strategies in tumor tissues using spatial transcriptomics
- 10.2025 **NYC Preparedness & Recovery Institute Narrative 2 Numbers (N2N) Hackathon**, NYC PRI, *New York, NY*
Project Title: Accelerating Case Definition and Outbreak Detection Through Community-Level Data Mining.
Developed a proactive disease surveillance pipeline leveraging unstructured, community-level data for rapid disease detection and phenotype definition. Awarded the Public Health Impact Prize and named runner-up with an AWS prize of \$1,000.
- 06.2015 – 08.2015 **Associate Intern**, Korea Railroad Research Institute, *South Korea*
Conducted data collection and analysis for research activities, and supported resolving technical issues and maintaining accurate laboratory records and reports.
- 06.2013 – 07.2013 **Laboratory Intern**, Department of Chemical Engineering, Sungkyunkwan University, *South Korea*

Supported laboratory research on solar and hydrogen energy by assisting with photocatalytic water-splitting experiments, collecting results, and organizing data.

07.2012 – 08.2012 **Laboratory Intern**, Department of Biochemistry, Yonsei University, *South Korea*
Assisted in gene-targeting research by producing and analyzing genetically modified mice and performing DNA extraction, mini prep, and midi prep using standard analytical equipment.

Work Experience

04.2021 – 05.2024 **Senior Scientist, Quality Assurance**, Samsung Biologics, *South Korea*
Qualified and released new API manufacturing facilities in Plant 4, accelerating readiness by more than a year.
Led validation lifecycle for a new mRNA manufacturing facility, shortening timeline by 2 months through modified verification strategies.
Cleaning validation: established cleaning limits for drug substances based on toxicity, degradation, and TOC recovery, minimizing cross contamination and ensuring quality and safety.
Risk management: reviewed and approved 20 “Deviation” records to capture policy violations 10 “Change Control” records annually to improve compliance and operational efficiency.

07.2017 – 04.2021 **Scientist, Validation**, Samsung Biologics, *South Korea*
Computerized System Validation: Supported verification of automated systems such as Manufacturing Control System (MCS) and Building Management System (BMS).
Equipment Validation: Reviewed and approved 30+ verification protocols/reports and implemented centralized laboratory system (LabX).
Data Integrity: Deployed backup servers and security controls to support compliance with 21 CFR Part 11 and Annex 11.
Technology Transfer: Assessed allowable TOC residue for 10 newly introduced products.

Selected Courseworks Projects

05.2025 **BIST P8106: Data Science II**, Columbia University, *New York, NY*
Project: Analysis on antibody response to vaccine and factors associated with incidence of flu.
Conducted predictive modeling of severe flu incidence post-vaccination using boosting, SVM, and interpretable methods, developing risk scores and identifying key demographic and clinical predictors.

04.2025 **BIST P8158: Latent Variable and Structural Equation Modeling**, Columbia University, *New York, NY*
Project: Review of mental health practitioners’ perceptions of AI analyzed via CFA and SEM.
Critiqued and evaluated a published study on mental health practitioners’ perceptions of AI using confirmatory factor analysis (CFA) and structural equation modeling (SEM).

12.2024 **BIST P8105: Data Science I**, Columbia University, *New York, NY*
 Project: Analysis of factors on Global Violence.
 Conducted a full data science pipeline on a real-world dataset, including data cleaning, exploratory visualization, formal analysis, and dissemination via a Shinyapp website and screencast.

Teaching Experience

01.2026 – 05.2026 **Teaching Assistant**, Columbia University, *New York, NY*
 Course: BISTP8180 Relational Databases and SQL Programming.
 Assisted with homework and exam grading, and supported in-person class sessions.

05.2022 – 01.2023 **Qualified Trainer for Validator**, Samsung Biologics, *South Korea*
 Provided comprehensive training on thermal data collection equipment validator to internal employees and supported troubleshooting to resolve common operational issues.

08.2018 – 08.2020 **Internal Instructor**, Samsung Biologics, *South Korea*
 Conducted classroom training on validation protocols for operators, providing in-depth knowledge of purpose, scope, testing methods and procedures in compliance with data integrity.

01.2016 – 05.2016 **Peer Tutor**, Cornell University, *Ithaca, NY*
 Peer-tutored freshmen and sophomores in general chemistry.

Extracurricular Activities

07.2025 – 09.2025 **Participant**, Columbia ISSO GLASS Program, Columbia University, *New York, NY*
 Completed Global Leadership Tier 3 Capstone Project through Columbia's International Students and Scholars Office, focusing on culturally competent mentorship.

09.2022 – 03.2024 **Secretary**, Over the Wall (Bouldering Club), Samsung Biologics, *South Korea*
 Coordinated monthly club activities and managed organizational budget.

07.2016 – 08.2016 **Instructor**, Gotoco, *Qingdao, China*
 Designed and delivered English as a second language lessons for high school students.

12.2015 – 01.2016 **Tutor & Mentor**, Project Teach, *Seoul, South Korea*
 Provided academic tutoring and mentorship for low-income students.

2014 – 2016 **Member**, Cornell Shimtah (Korean Traditional Music Club), Cornell University, *Ithaca, NY*
 Performed in annual concerts as a member of a Korean traditional music group.

2015 – 2016 **President**, Cornell 4C (Catholic Community Club), Cornell University, *Ithaca, NY*
 Coordinated community service activities and cultural events.

Certificates & Skills

Software and Programming: R Studio, Python, Microsoft Office, MySQL, LaTeX

Languages:

Fluent in Korean and English

Conversational in French (DELF/DALF B1, May 2020)

Certificates:

Columbia ISSO Global Leadership Tier 1 (06.2025)

Data Integrity Training (Nov 2019), Cleaning Validation (03.2023)

Coursera: Chemical Biology from University of Geneva (04.2023)

Coursera: Python Data Structures from University of Michigan (04.2021)

Coursera: Programming for Everybody from University of Michigan (02.2021)